NOTE An ESA SHALL always use the same value to represent the same operating state.

By providing this information a smart EMS may be able to learn the observed power draw when the ESA is put into a specific state. It can potentially then use the ManufacturerESAState field in the Forecast attribute along with observed power drawn to predict the power draw from the appliance and potentially ask it to modify its timing via one of the adjustment request commands, or adjust other ESAs power to compensate.

9.2.7.14.10. NominalPower Field

This field SHALL indicate the expected power that the appliance will use during this slot. It may be considered the average value over the slot, and some variation from this would be expected (for example, as it is ramping up).

9.2.7.14.11. MinPower Field

This field SHALL indicate the lowest power that the appliance expects to use during this slot. (e.g. during a ramp up it may be 0W).

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative power.

9.2.7.14.12. MaxPower Field

This field SHALL indicate the maximum power that the appliance expects to use during this slot. (e.g. during a ramp up it may be 0W). This field ignores the effects of short-lived inrush currents.

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative power.

9.2.7.14.13. NominalEnergy Field

This field SHALL indicate the expected energy that the appliance expects to use or produce during this slot.

Some appliances (e.g. battery inverters which can charge and discharge) may have a negative energy.

9.2.7.14.14. Costs Field

This field SHALL indicate the current estimated cost for operating.

For example, if the device has access to an Energy pricing server it may be able to use the tariff to estimate the cost of energy for this slot in the power forecast.

When an Energy Management System requests a change in the schedule, then the device MAY suggest a change in the cost as a result of shifting its energy. This can allow a demand side response service to be informed of the relative cost to use energy at a different time.

The Costs field is a list of CostStruct structures which allows multiple CostTypeEnum and Values to be shared by the energy appliance. These could be based on GHG emissions, comfort value for the